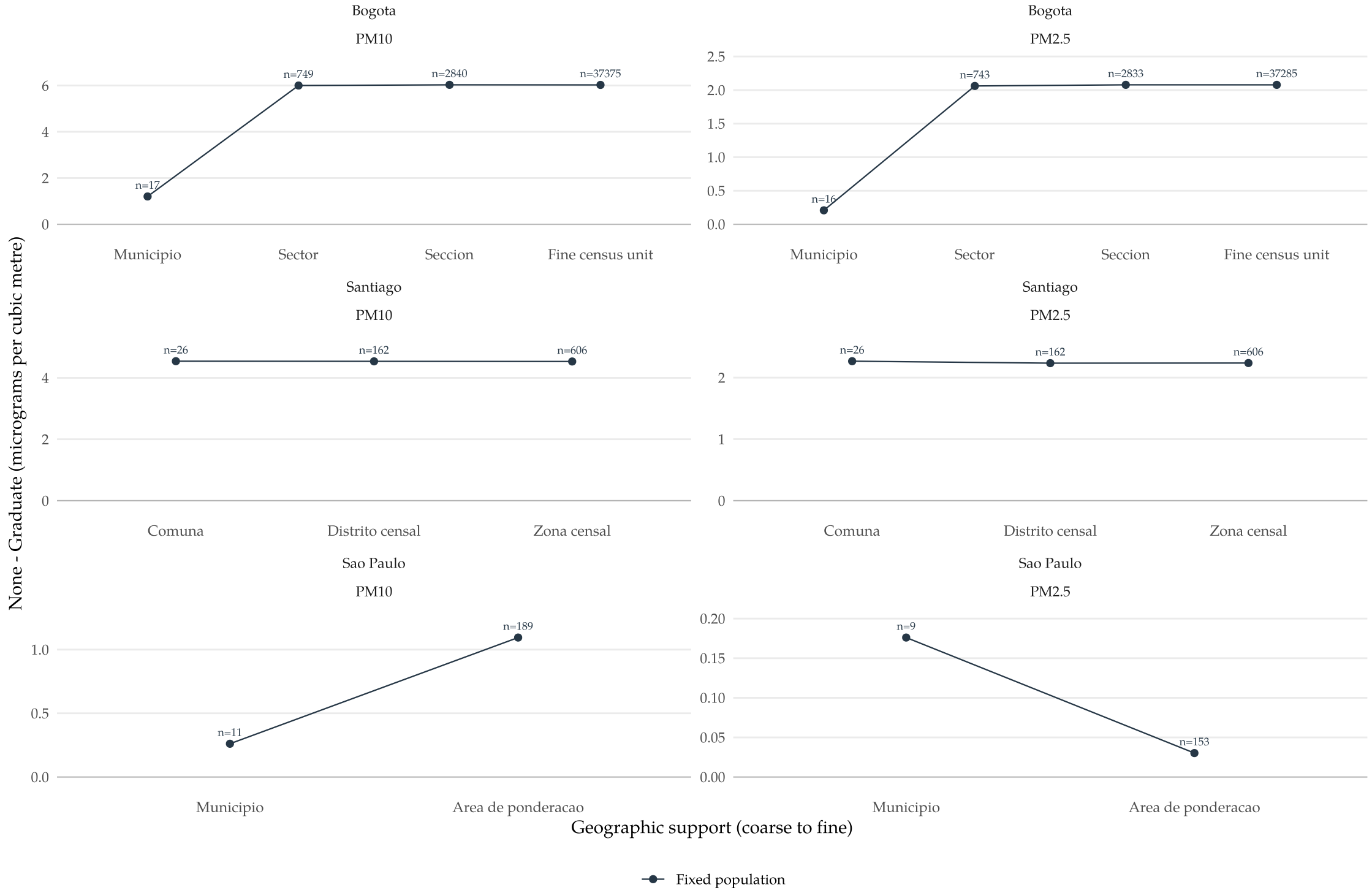
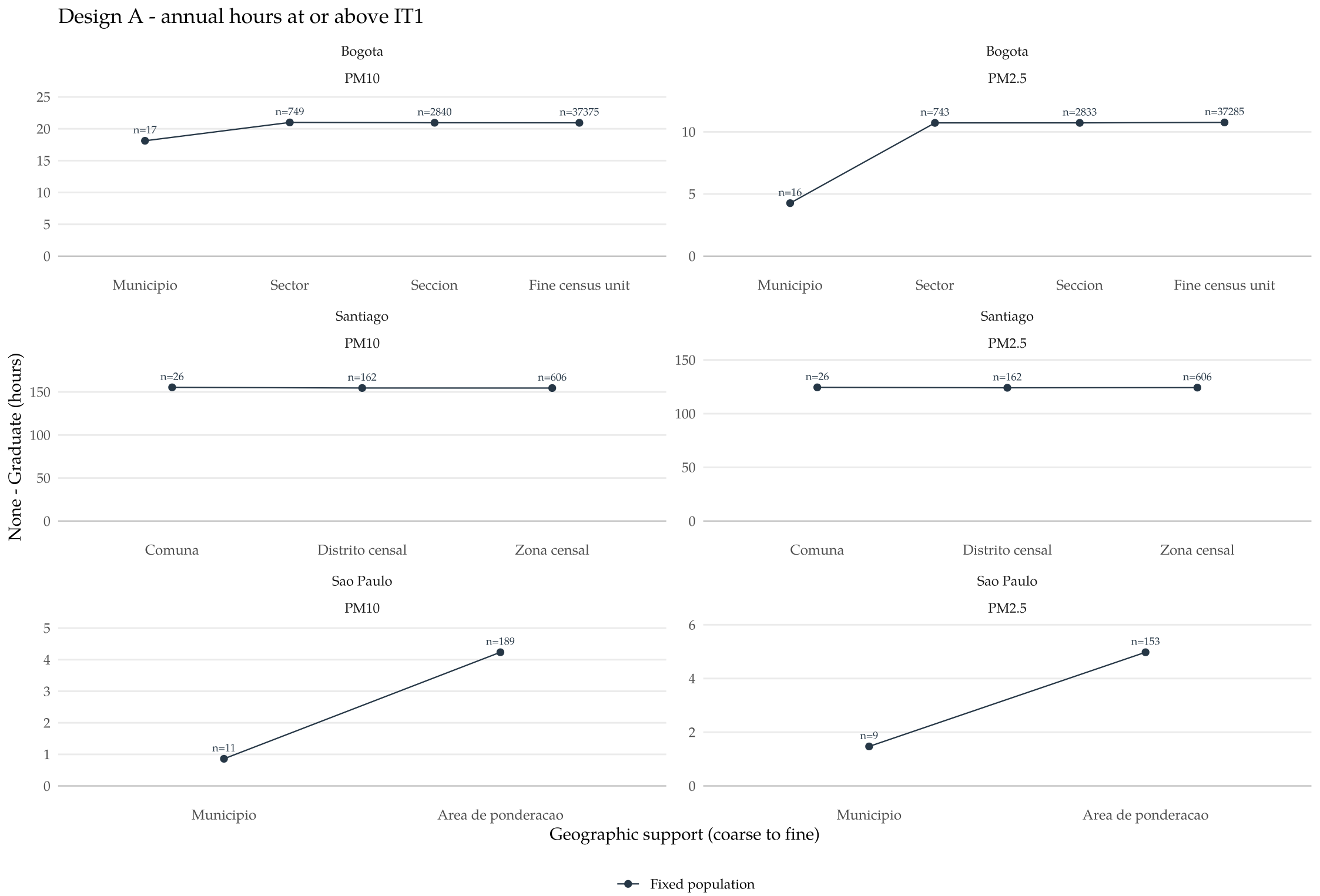
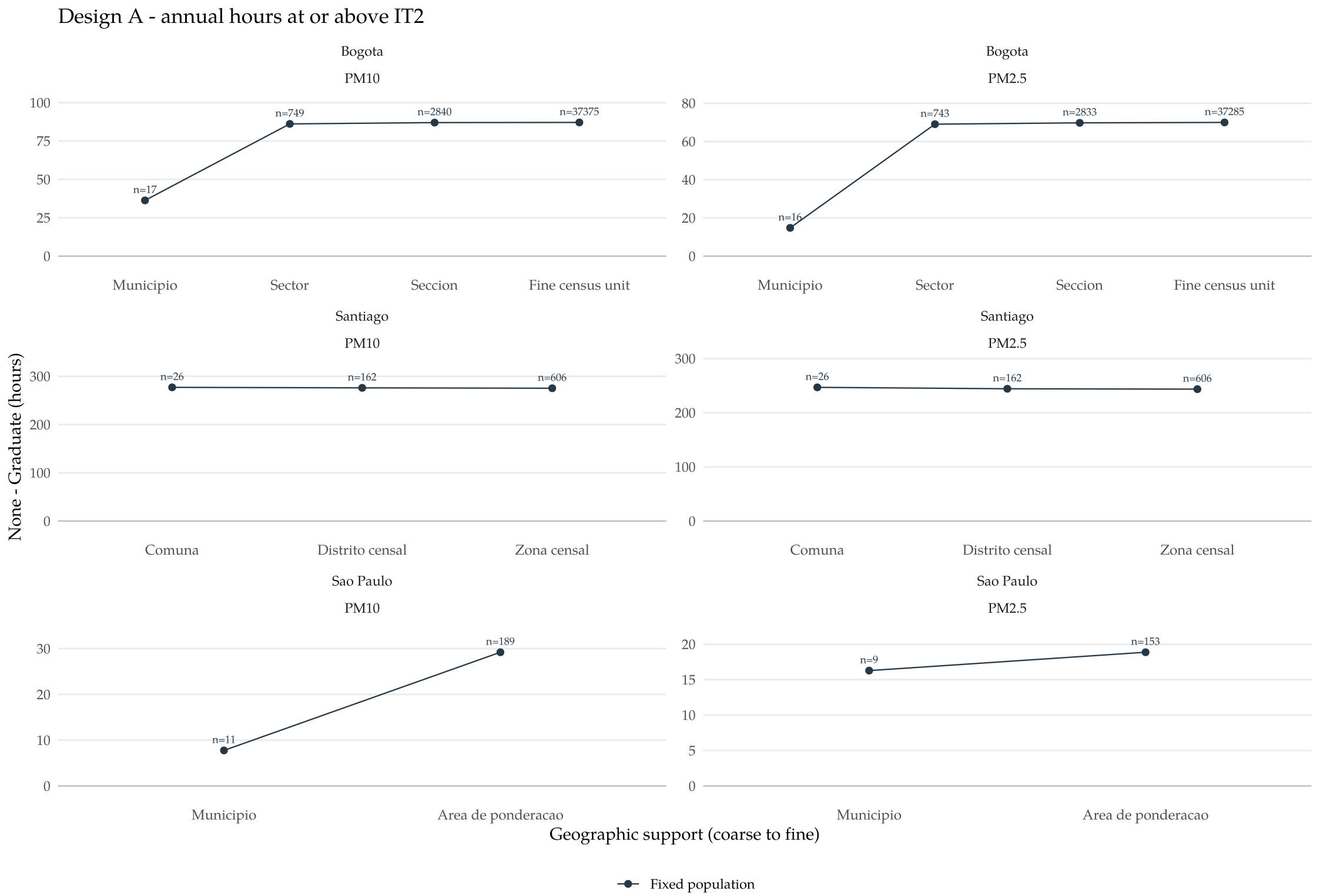
Design A - annual mean exposure



Fixed education-reporting adults and individual levels. Exposure aggregated with all-adult weights; locality branch shown separately.

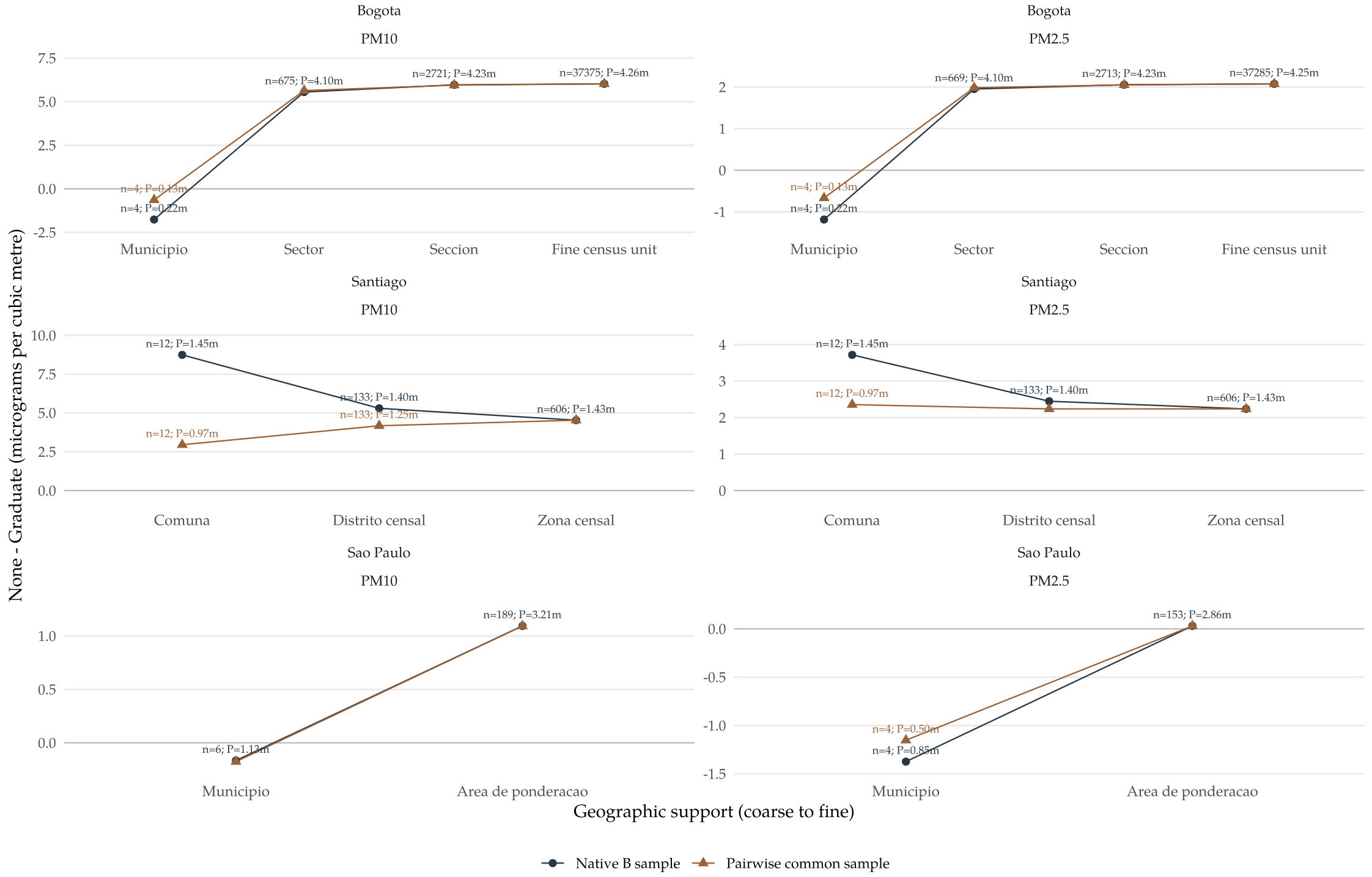


Fixed education-reporting adults and individual levels. Exposure aggregated with all-adult weights; locality branch shown separately.



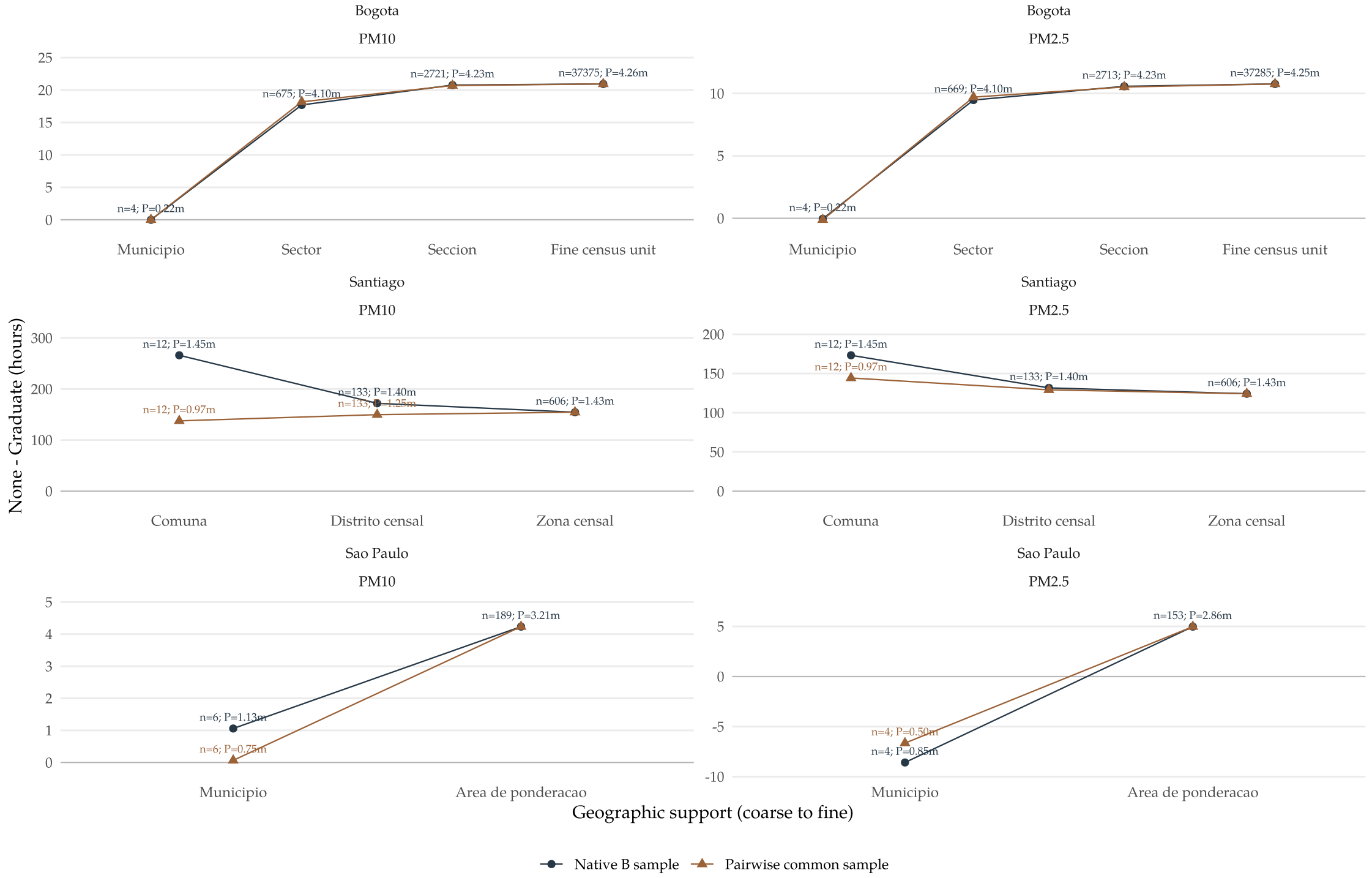
Fixed education-reporting adults and individual levels. Exposure aggregated with all-adult weights; locality branch shown separately.

Design B - annual mean exposure



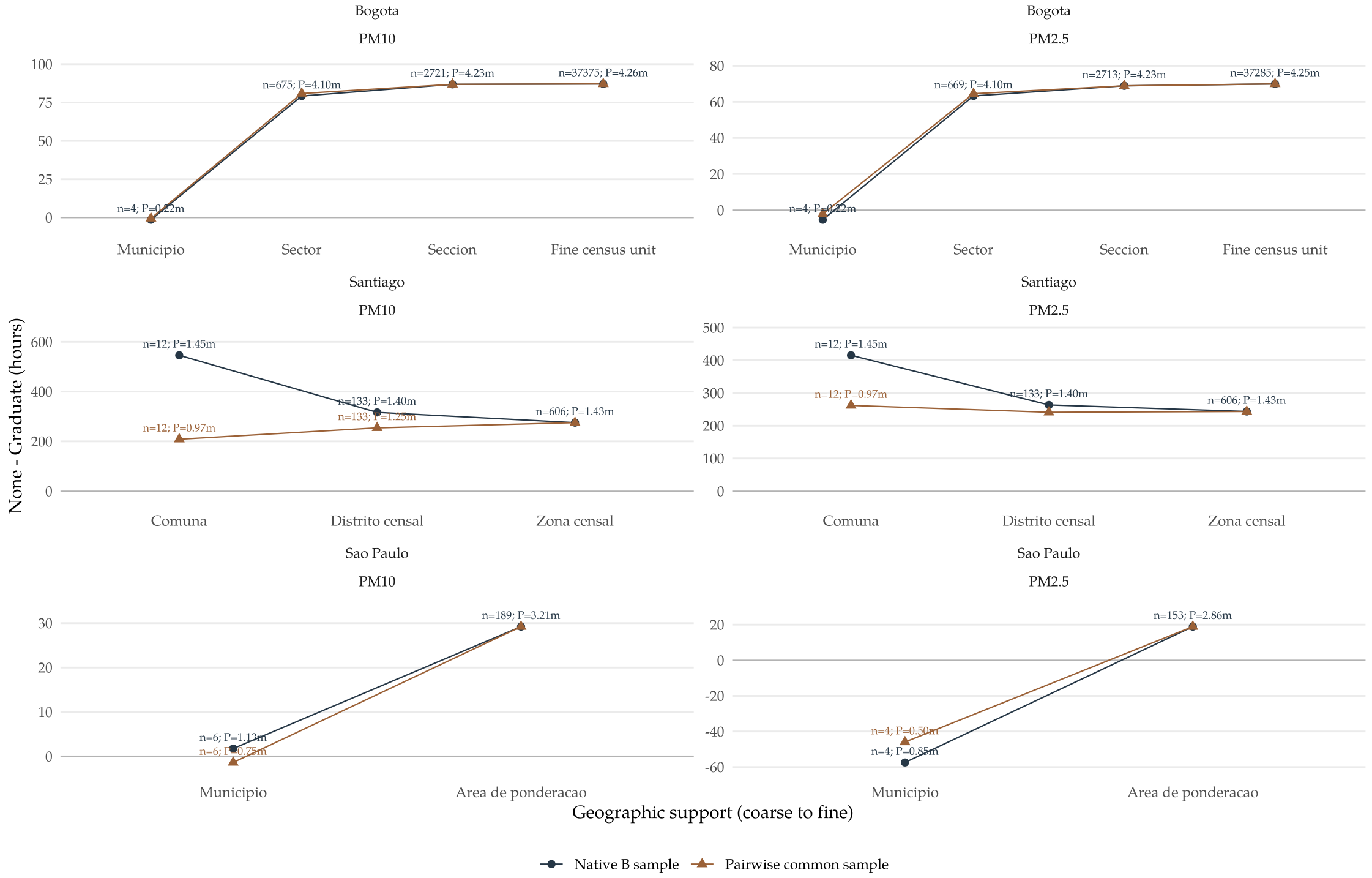
Native and common samples vary by level; P = education-reporting adults.
Common samples are pairwise intersections. Paired changes are shown separately.

Design B - annual hours at or above IT1

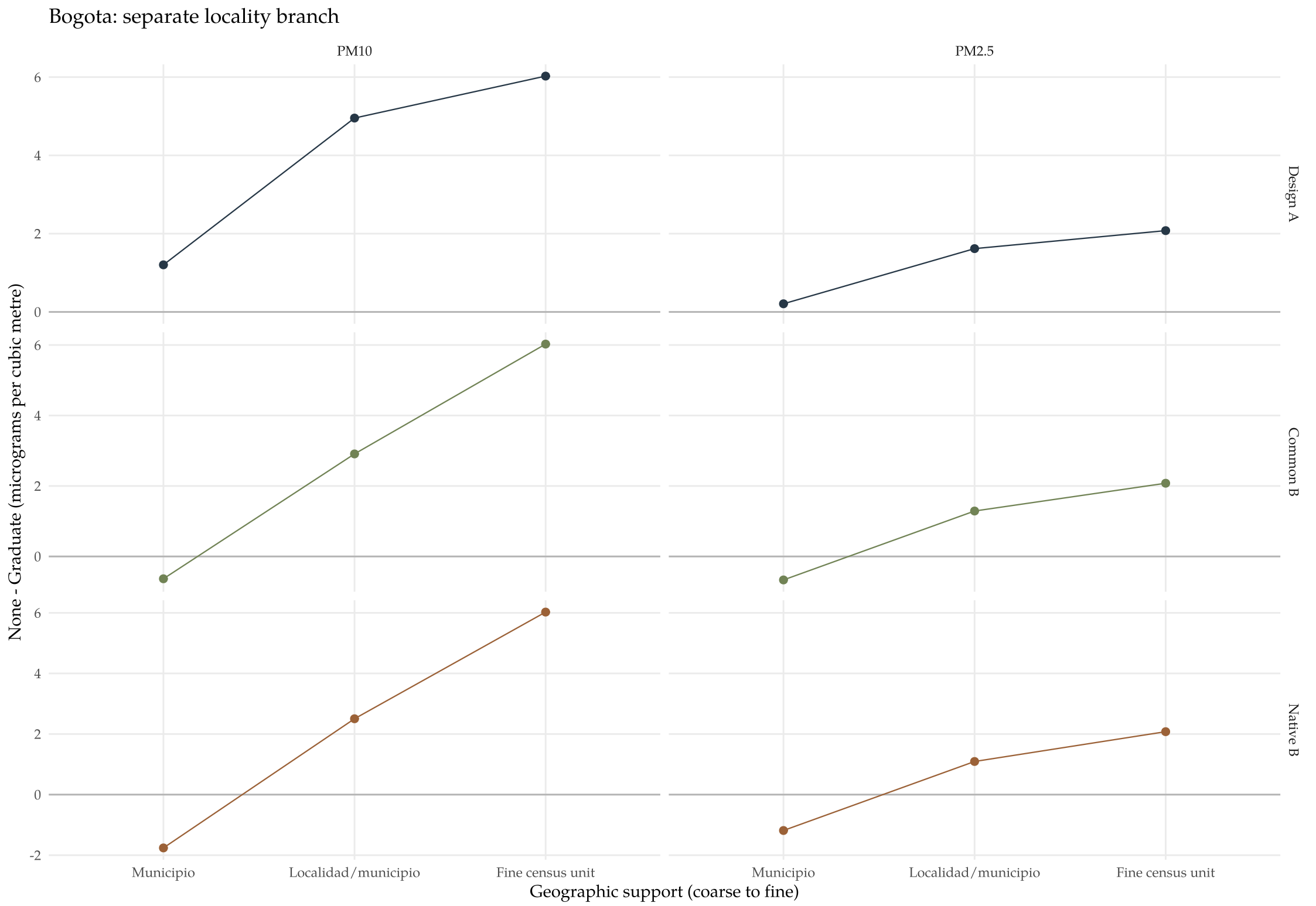


Native and common samples vary by level; P = education-reporting adults.
Common samples are pairwise intersections. Paired changes are shown separately.

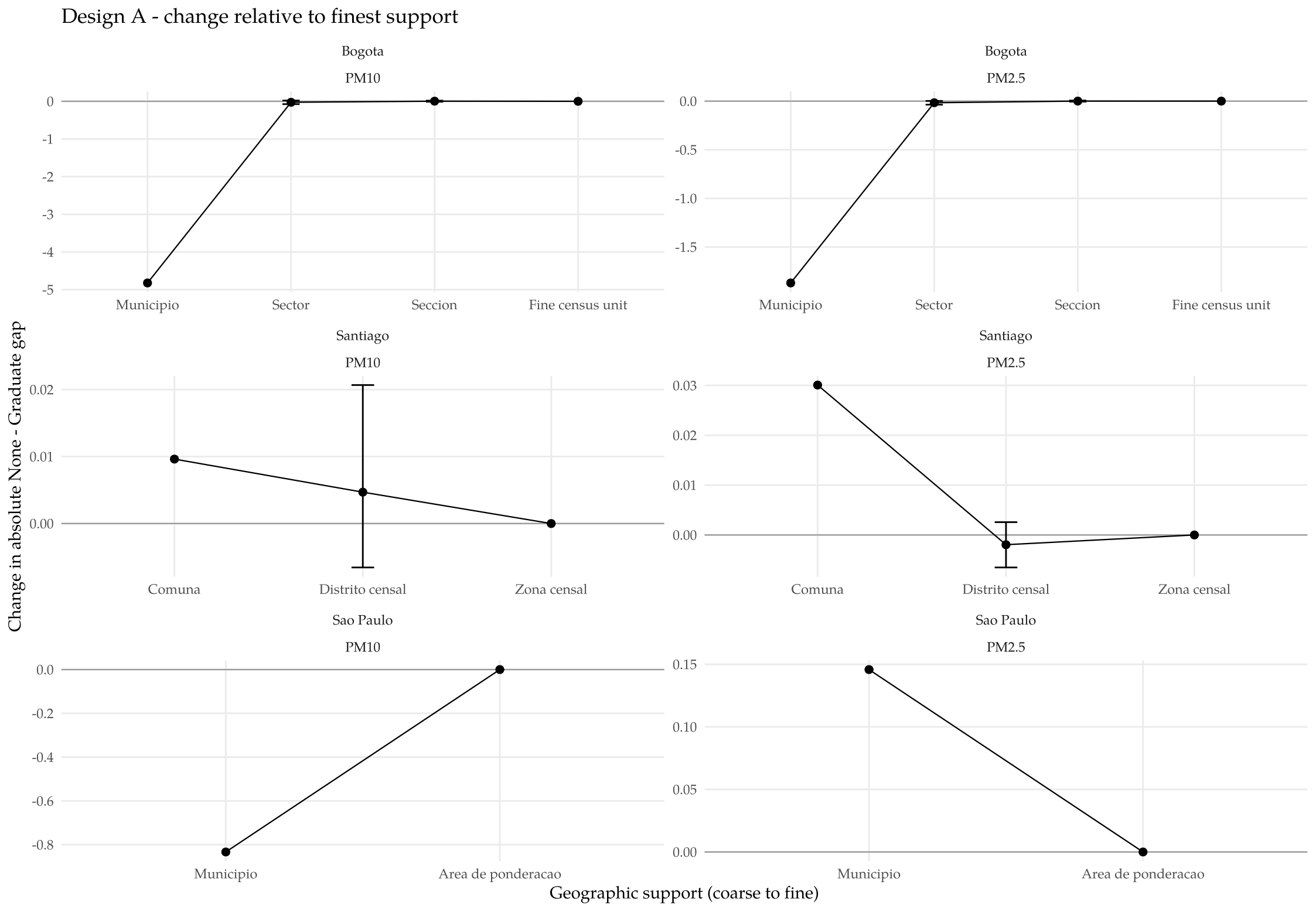
Design B - annual hours at or above IT2



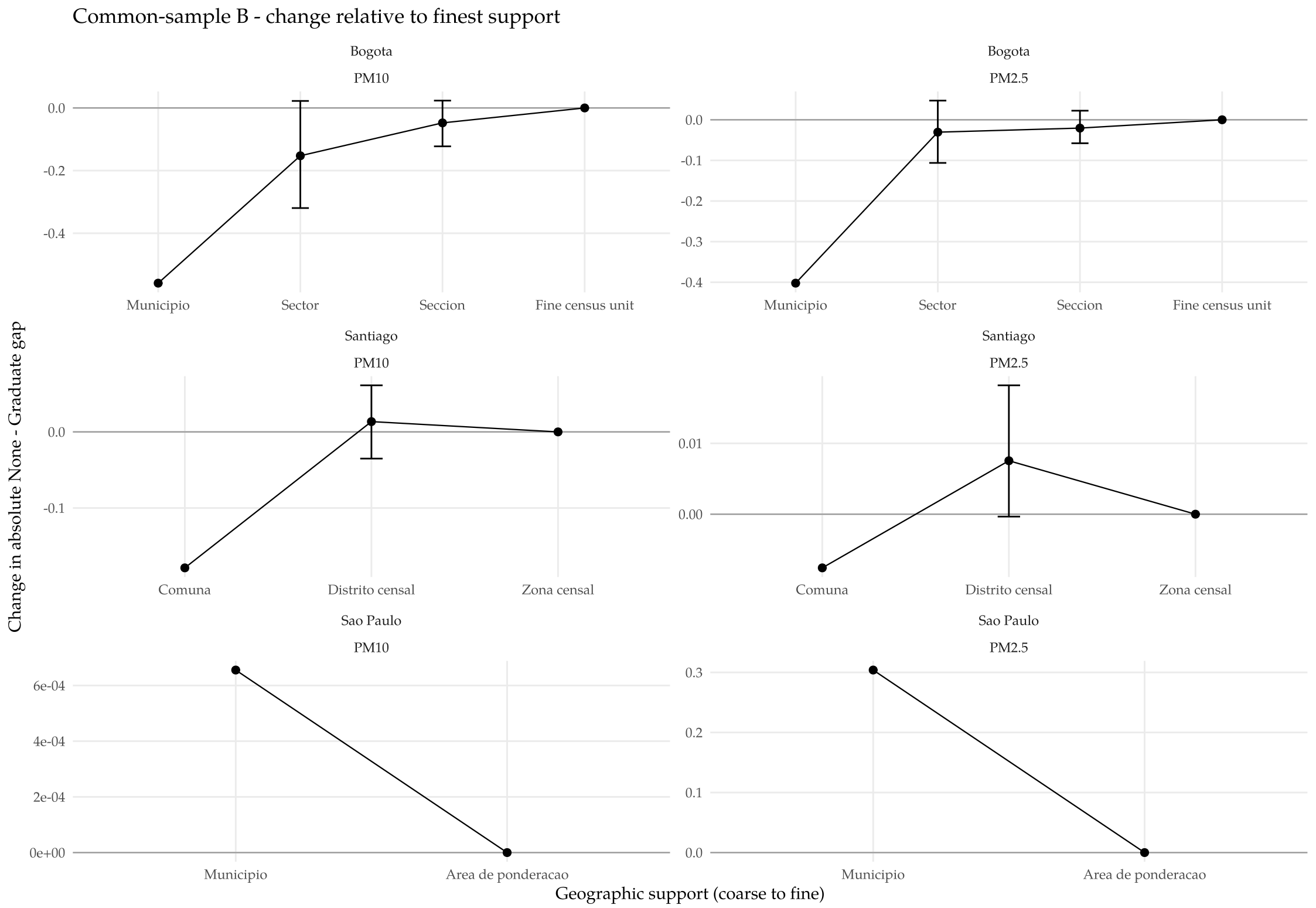
Native and common samples vary by level; P = education-reporting adults.
Common samples are pairwise intersections. Paired changes are shown separately.



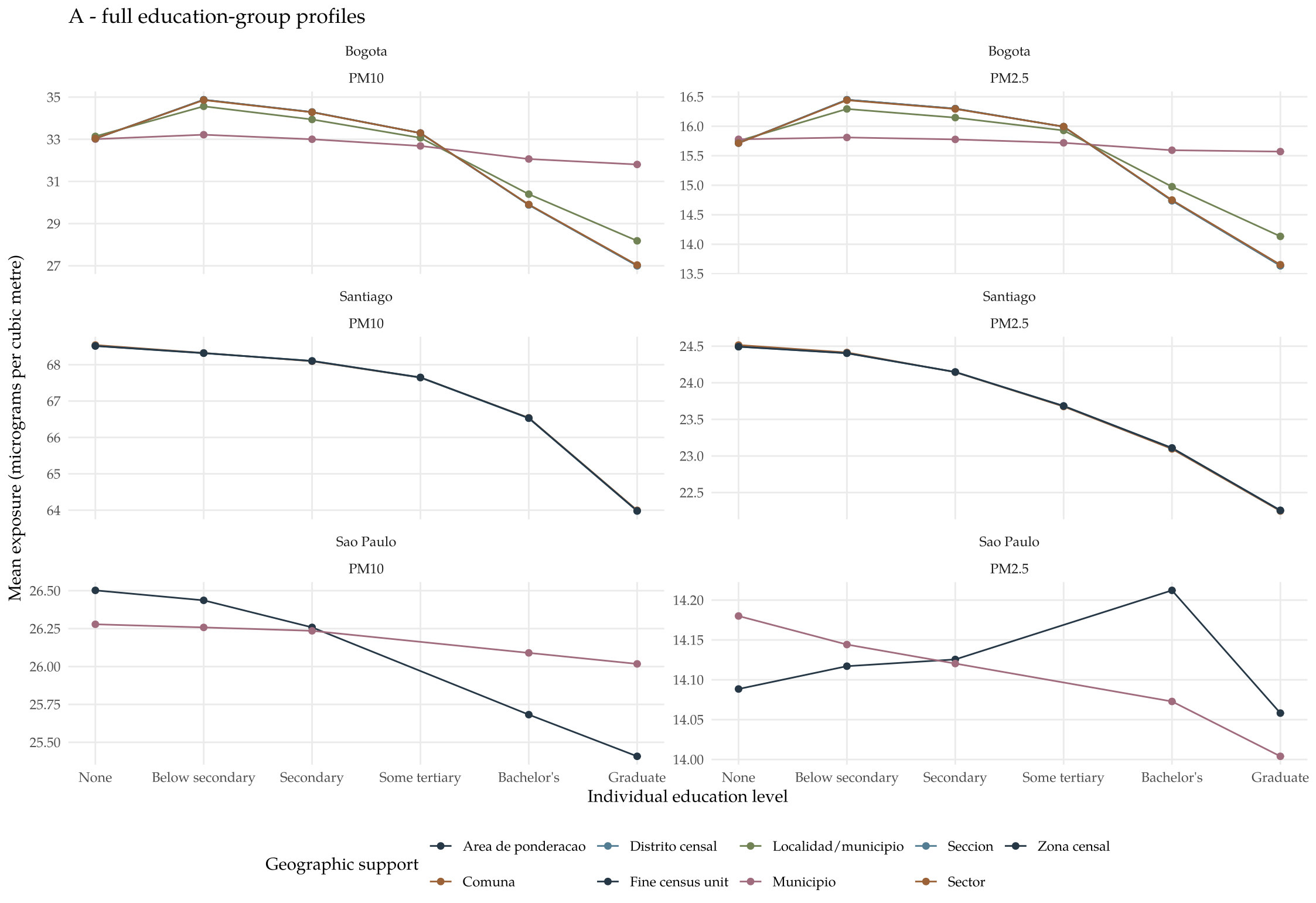
Descriptive comparisons.



95% conditional geographic-cluster bootstrap intervals; 999 draws. Coarse comparisons are descriptive. B endpoints use the same pairwise population.

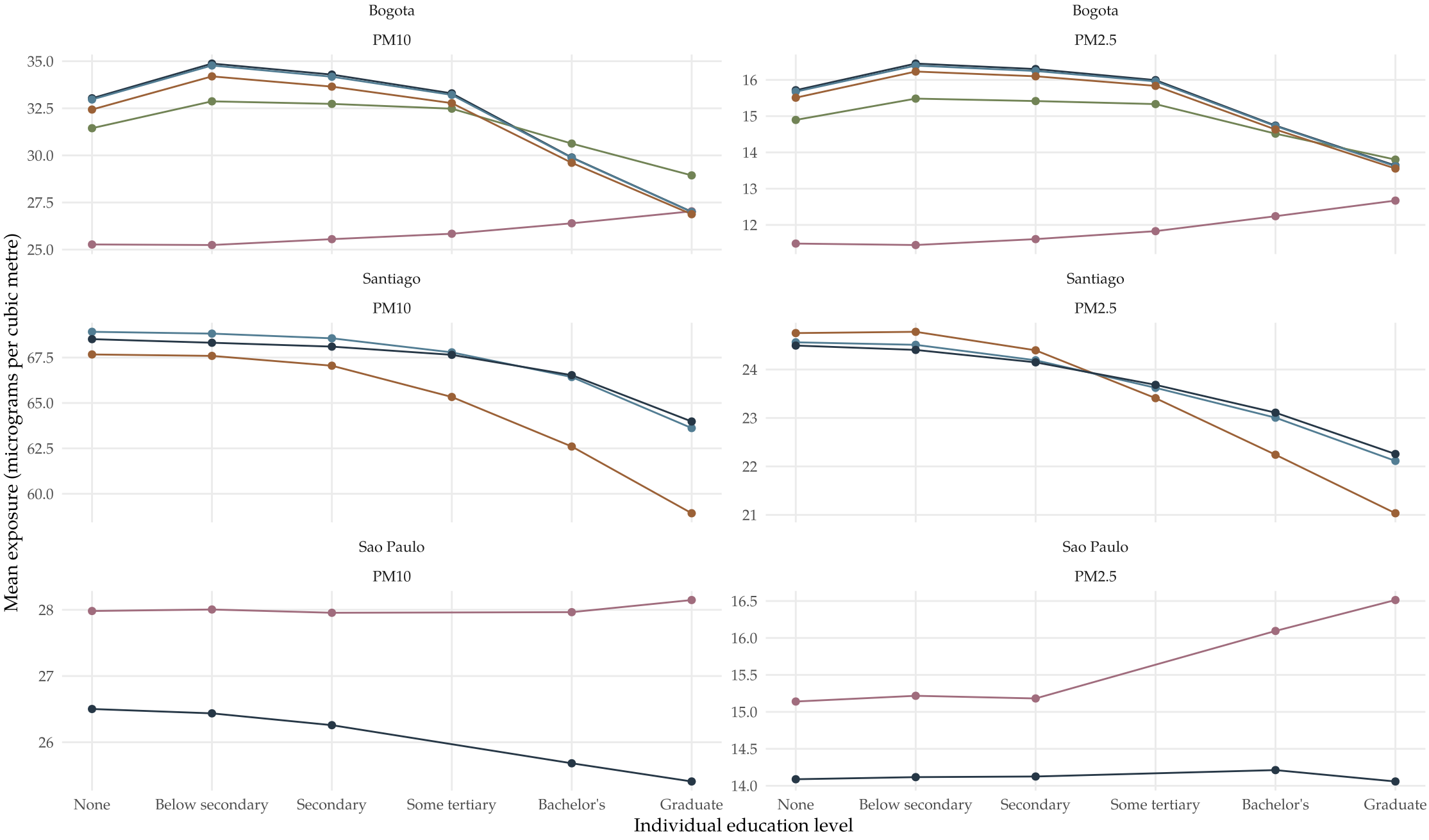


95% conditional geographic-cluster bootstrap intervals; 999 draws. Coarse comparisons are descriptive. B endpoints use the same pairwise population.



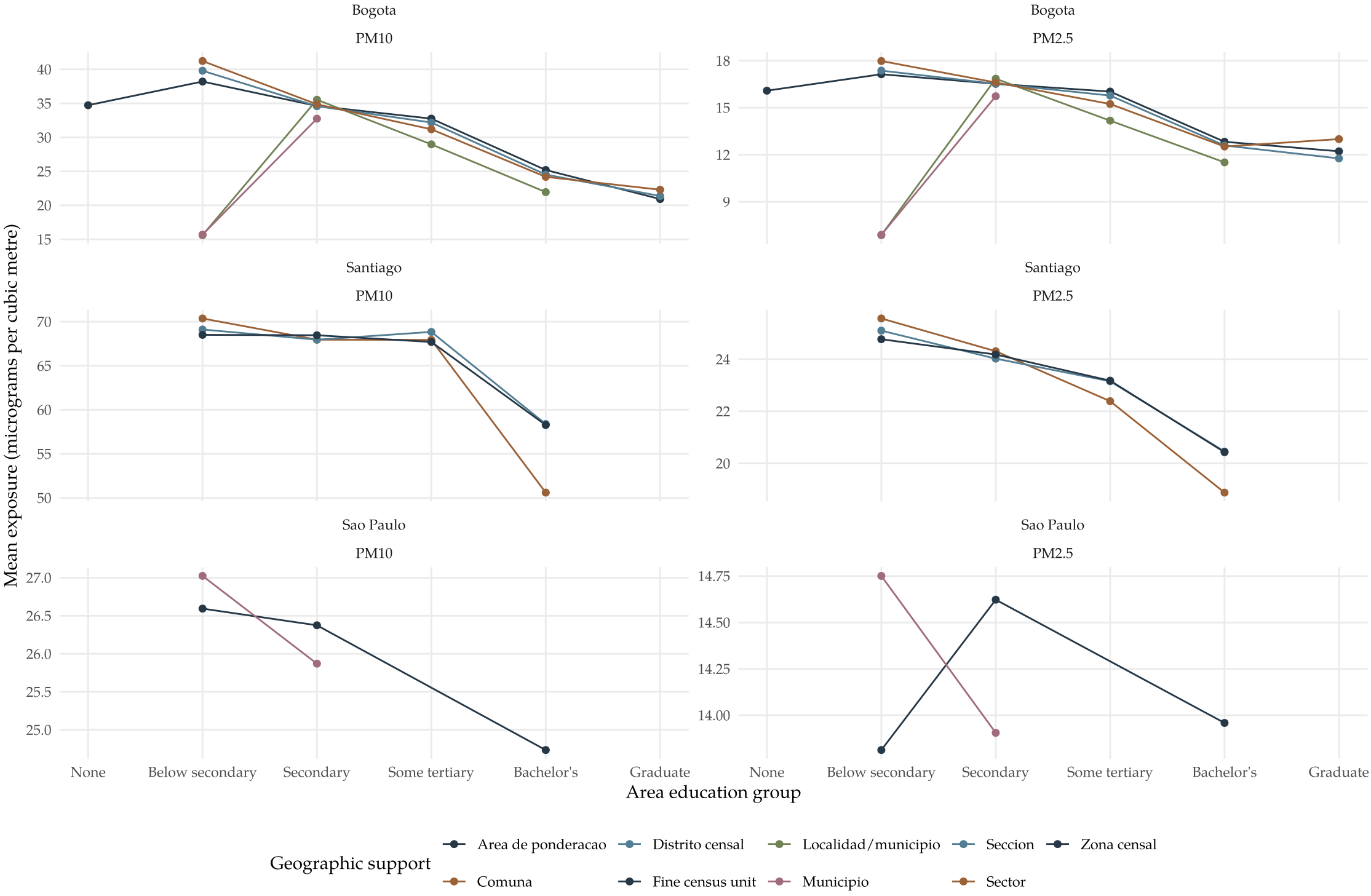
Harmonized education levels; group means use existing person weights. Sao Paulo's census cannot code 'Some tertiary'. Bogota's 'Some tertiary' includes completed tecnica and normalista.

Native B - full education-group profiles



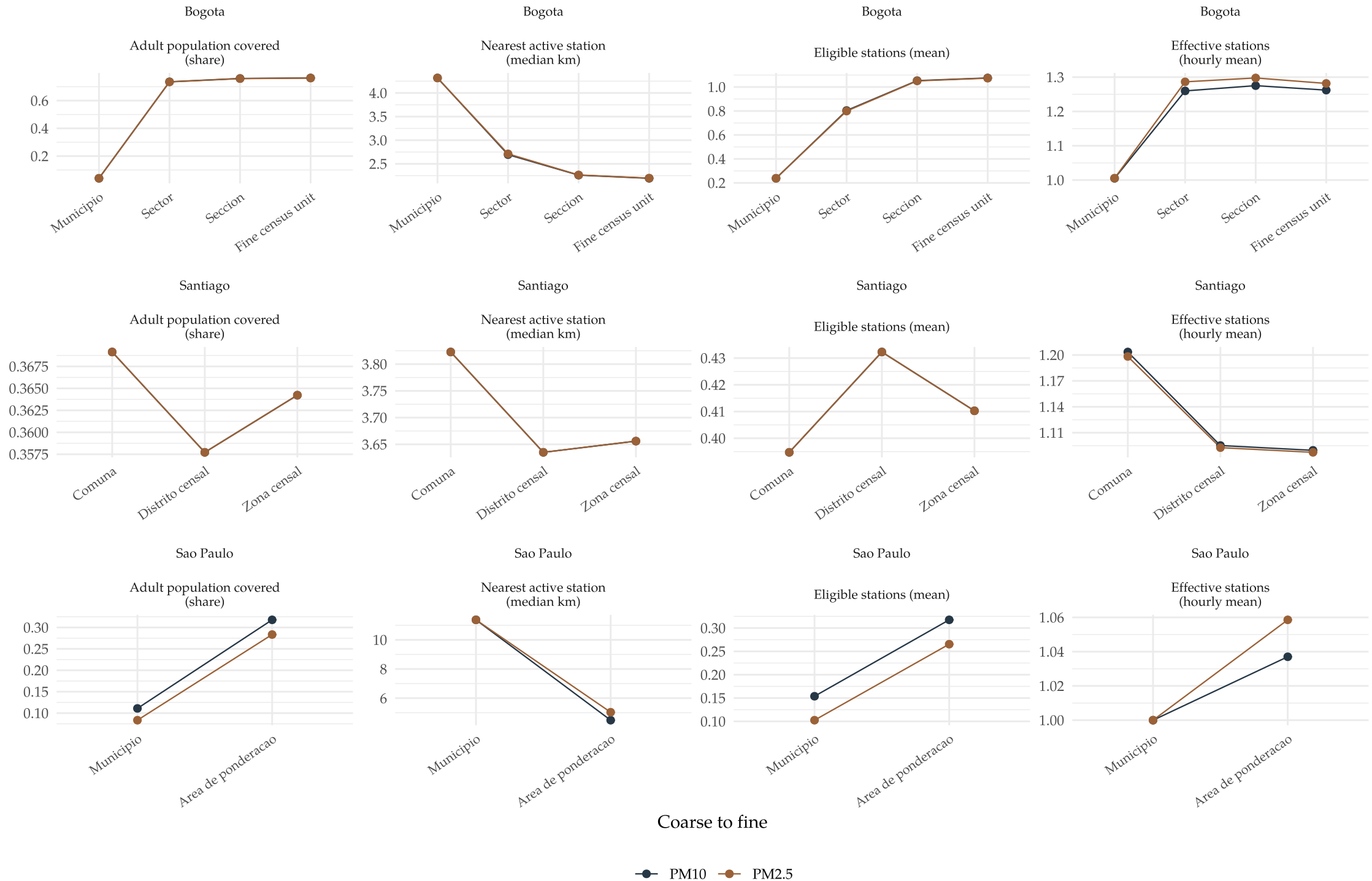
Harmonized education levels; group means use existing person weights. Sao Paulo's census cannot code 'Some tertiary'. Bogota's 'Some tertiary' includes completed tecnica and normalista.

C - full education-group profiles



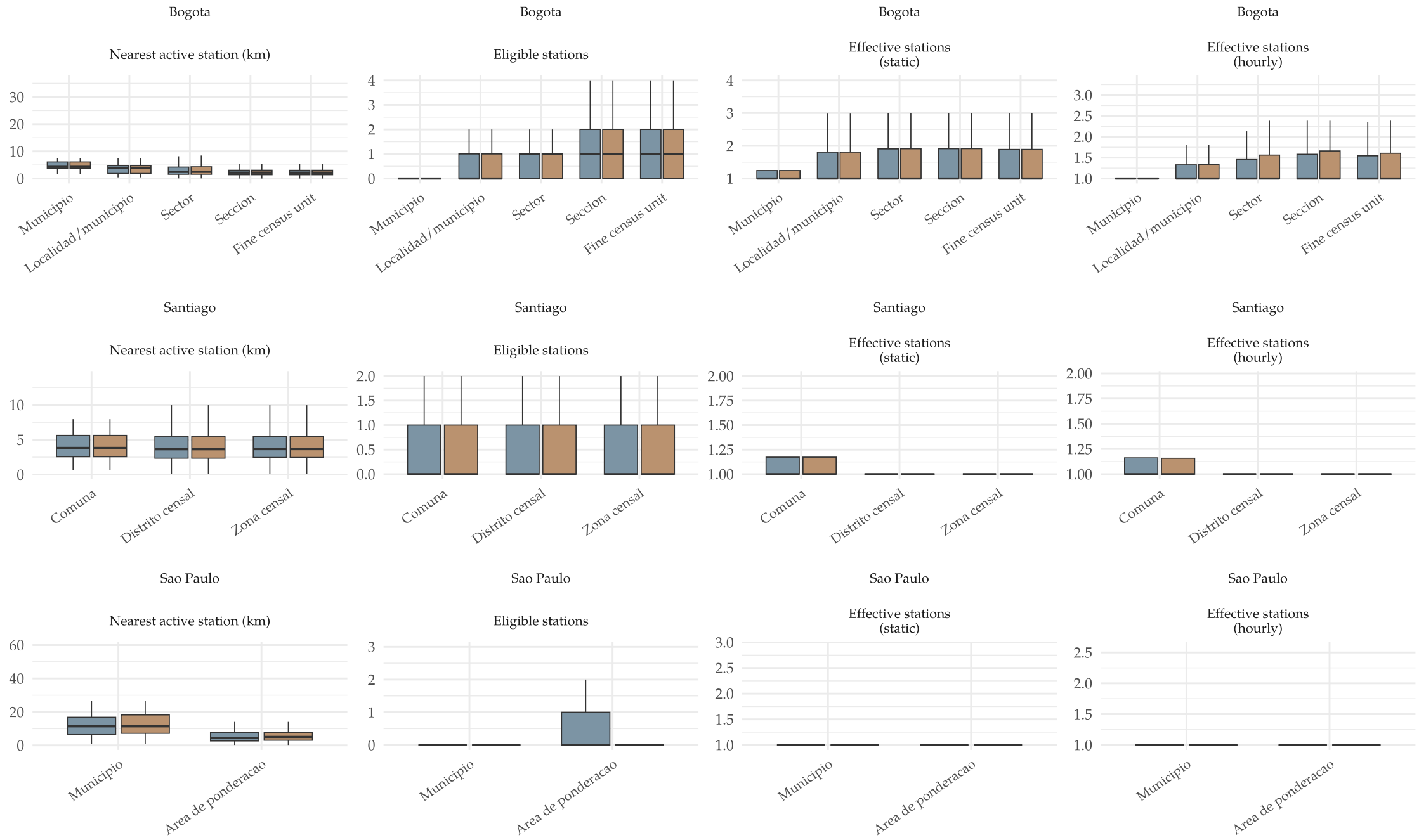
Classification only; finest exposure fixed. Empty area groups remain unavailable. These groups differ from A/B levels.

Monitoring-matrix sensitivity under Design B

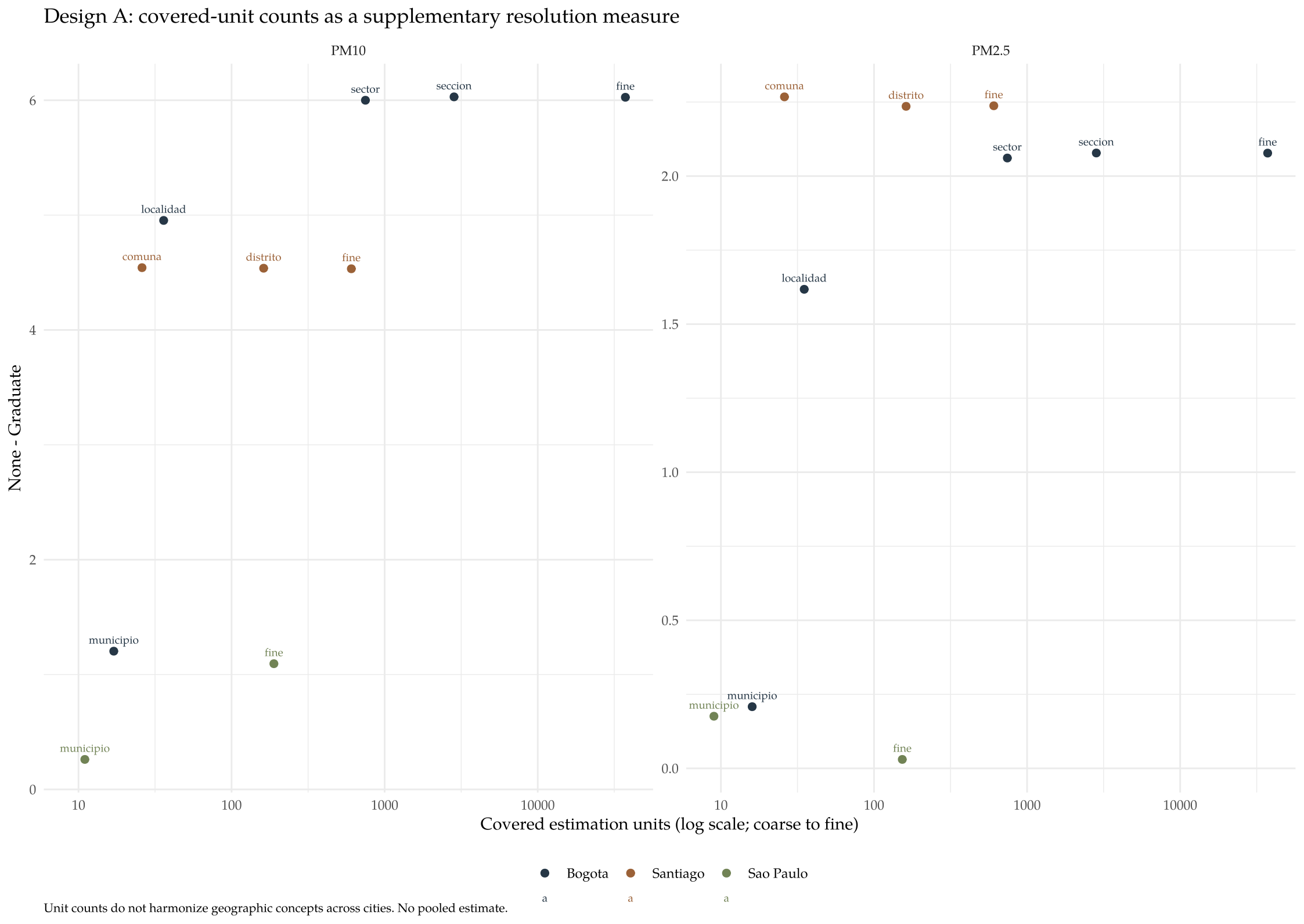


Catalog, active, eligible and contributing station counts are distinct in the tables.

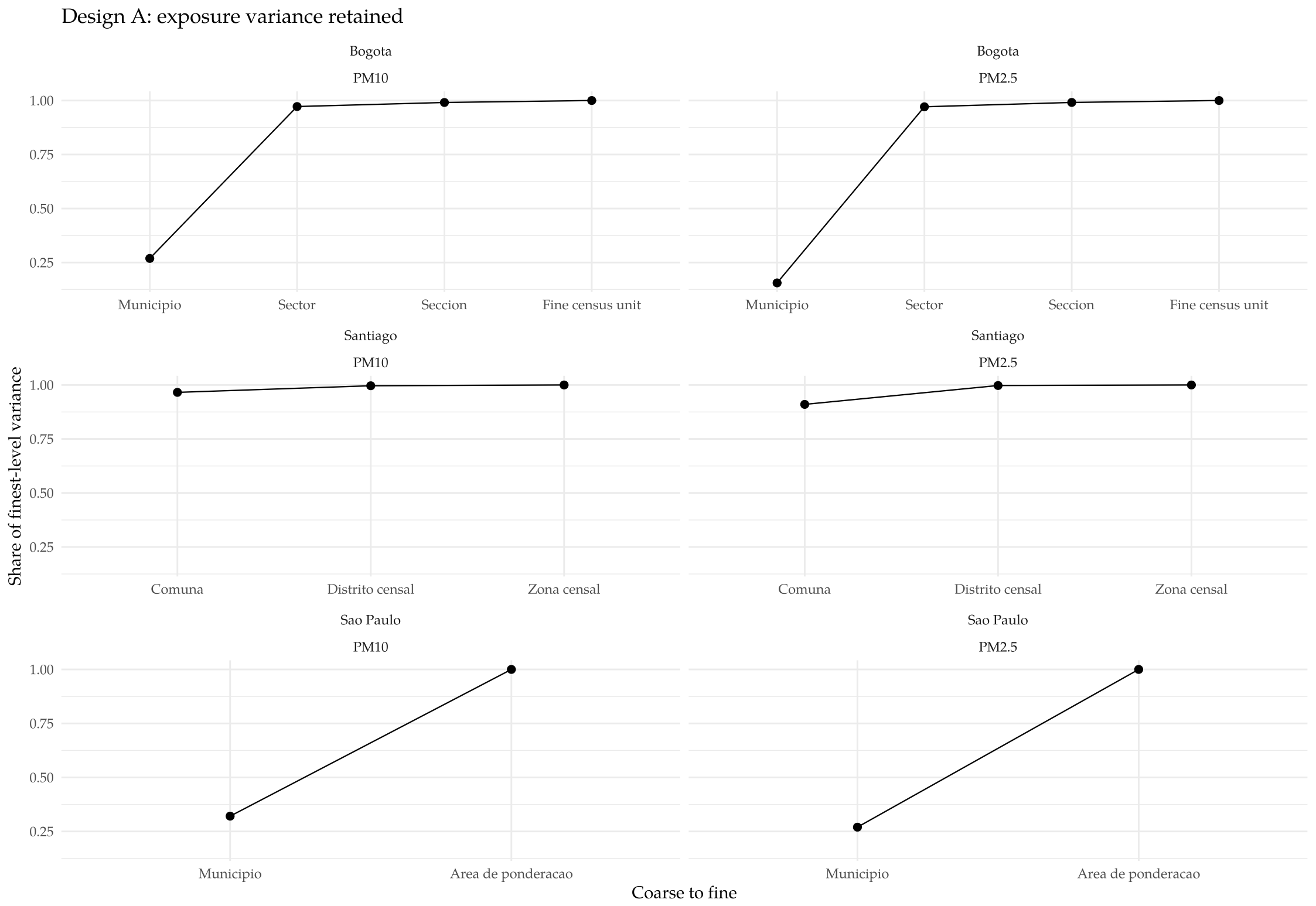
Monitoring-matrix distributions across populated units



Boxes: unweighted unit quartiles; whiskers: 1.5 IQR. Outliers omitted from view only.



Unit counts do not harmonize geographic concepts across cities. No pooled estimate.



All-adult aggregation weights and fixed covered support. Declining variance does not imply a monotone gap.